

Bilal Shihab

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PROFESSIONAL SUMMARY

Biomedical Engineering undergraduate specializing in Embedded Systems, Edge AI, and hardware-software integration. Passionate about developing low-level firmware and high-performance data pipelines to bridge the gap between hardware and software.

EDUCATION

UNIVERSITY OF TEXAS AT AUSTIN

Austin, TX

Bachelor of Science in Biomedical Engineering, Computational Track

May 2027

- **Relevant Coursework:** Embedded Systems, Biomedical Instrumentation, Circuits, Systems & Signals, Differential Equations & Linear Algebra, Intro to Computational Engineering Design, Statistics, Numerical Methods

SKILLS

- **Languages:** Python, C, ARMv6-M Assembly, MATLAB, R
- **Embedded & Hardware:** TI MSPM0, Raspberry Pi, Arduino, Code Composer Studio, UART, SolidWorks
- **AI & Data:** PyTorch, ExecuTorch, Captum, Optuna, Google Cloud, TensorFlow Lite, Jupyter Notebook

EXPERIENCE

LONGHORN NEUROTECH

Austin, TX

Edge AI Team Lead

Sep 2025 - Present

- Directed a team of student engineers to optimize neural networks for deployment on embedded systems (Raspberry Pi series).
- Designing a reusable quantization pipeline using ExecuTorch and an inference backend to support keyword spotting and future neuroprosthetic control applications.
- Developing a low-latency Python audio acquisition module for USB audio devices, using circular buffering to stream live audio into an on-device inference engine.

AI/ML Developer

Sep 2024 - Aug 2025

- Developed machine learning models for Brain-Computer Interfaces (BCI) aimed at controlling a prosthetic arm, using architectures such as CNNs and Capsule Networks.
- Improved BCI model accuracy from 60% to 80% by tuning hyperparameters with Optuna and using Captum to reduce overfitting.

RECANZONE LABORATORY, UC DAVIS

Davis, CA

Research Intern

Jun 2025 - Aug 2025

- Developed an interactive 3D feature visualization UI for open-source neuroscience software, allowing users to inspect neural features through real-time camera and projection controls.
- Implemented real-time rotation and transformation logic using linear algebra, enabling mouse-based inspection of neural spike data.

FUNCTIONAL OPTICAL IMAGING LAB, UT AUSTIN

Austin, TX

Researcher and Programmer

Sep 2023 - May 2025

- Developed a real-time Laser Speckle Analysis UI in Python, interfacing with Basler cameras and Arduinos for automated laser intensity control.
- Enhanced speckle pattern evaluation by implementing contrast analysis algorithms to maximize measurement precision.
- Refactored contrast adjustment scripts from MATLAB to Python, enabling integration with the real-time Laser Speckle Analysis UI.

TECHNICAL PROJECTS

CLOUD TO EDGE HEART RATE MONITOR

Dec 2025 - Present

- Architected a pipeline that retrieves heart rate datasets from Cloud Storage to a Raspberry Pi 5, utilizing it as a patient simulator.
- Developed a UART communication protocol to stream simulated patient data from the Pi 5 to a Pi Pico for low-latency processing.
- Built a custom C-based inference engine on the Pico's dual-core Cortex-M0+ architecture to execute quantized neural networks on the incoming data stream in order to classify the patient's heart condition.

EMBEDDED MULTIPLAYER RACER (ARM CORTEX-M0+)

Nov 2025 - Dec 2025

- Earned 'Best Embedded Design' in a class-wide competition for excellence in hardware-software integration.
- Built a bare-metal multiplayer racing game in C for a TI MSPM0 microcontroller, implementing collision detection, velocity-based physics, and sprite rendering.
- Programmed register-level drivers for UART, ADC, and DAC to interface with analog joysticks and audio peripherals, bypassing standard libraries for efficiency.
- Designed a custom UART protocol to synchronize player coordinates and game events between two microcontrollers in real-time.

ECG PCB PROJECT

Mar 2025 - Apr 2025

- Designed a custom Electrocardiogram PCB in Autodesk Fusion 360, creating the schematic for physiological monitoring.
- Engineered the analog signal chain, implementing instrumentation amplifiers and active filtering (bandpass/notch) to reject 60Hz noise and isolate cardiac signals.
- Coordinated component selection and fabrication logistics, optimizing the Bill of Materials for cost and assembly efficiency.